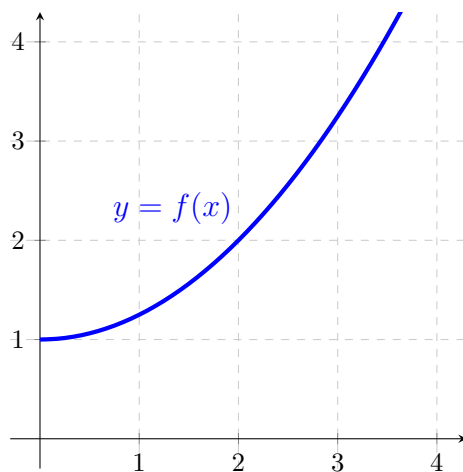


Problem 7

Problem 7

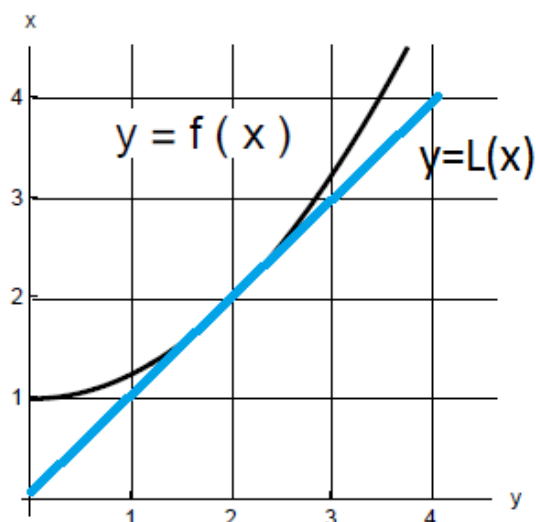
Problem 1 The graph of a function f is given below.



- (a) Given that $f'(2) = 1$, find the linear approximation L to the function f at $a = 2$. [2]

Explanation. $L(x) = f(2) + f'(2)(x - 2) = 2 + (x - 2) = x$

- (b) Sketch the graph of L in the figure above.

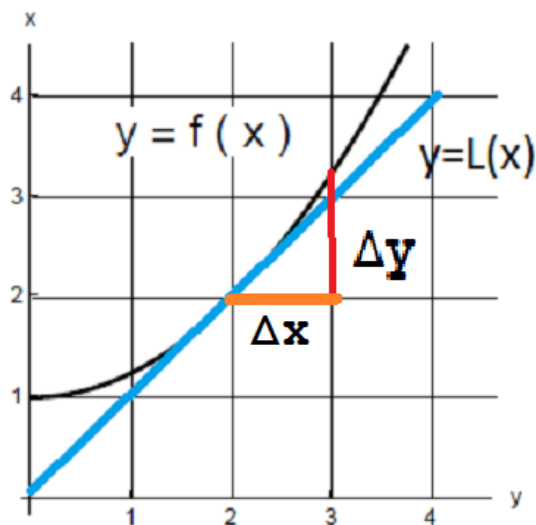


Explanation.

- (c) Use the linear approximation L to estimate the value of $f(3)$. Is this an underestimate or overestimate? **EXPLAIN.** [2]

Explanation. $f(3) \approx L(3) = 3$. It is an underestimate because f is concave up on the interval (2, 3).

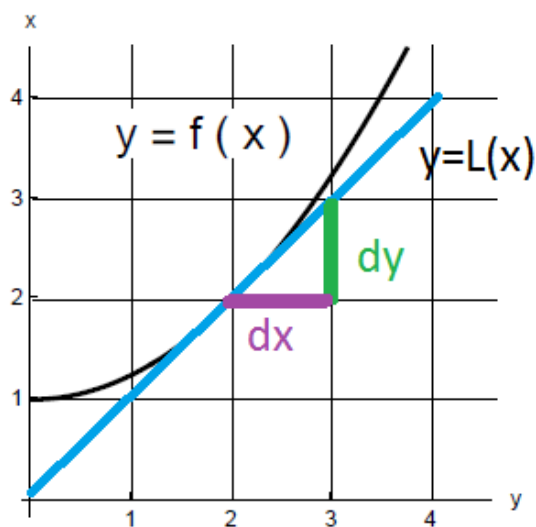
- (d) When x changes from $a = 2$ to $a + \Delta x = 3$, the change in the function $y = f(x)$, Δy , is given by $\Delta y = f(a + \Delta x) - f(a)$. Draw and label Δy and Δx in the figure above.



Explanation.

Problem 7

- (e) When x changes from $a = 2$ to $a + \Delta x = 3$, the change in the **linear approximation**, dy , is given by $dy = L(a + \Delta x) - L(a) = f'(a)\Delta x$. Draw and label $L(x)$, dx and dy (differential) in the figure above.



Explanation.